

Guided Notes: Module 4 – Configuring Network Addressing

Module Objectives

- Explain IPv4 addressing schemes.
- Explain IPv4 forwarding.
- Configure IP networks and subnets.
- Use tools to test a host's IP configuration.
- Explain IPv6 addressing schemes.
- Troubleshoot IP networks and hosts.

Lesson 4.1: Internet Protocol Basics

Key Concepts:

- **IPv4 Datagram Header:** Structure of the packet used in IPv4 communication.
- **Layer 2 vs Layer 3 Addressing:** Layer 2 uses MAC addresses; Layer 3 uses IP addresses.
- **Address Resolution Protocol (ARP):** Maps Layer 3 IP addresses to Layer 2 MAC addresses.
- **Unicast Addressing:** One-to-one communication.
- **Broadcast Addressing:** One-to-all communication within a network.
- **Multicast Addressing:** One-to-many communication to interested hosts using special IP ranges.
- **Anycast Addressing:** One-to-nearest communication, used by routers and protocols like BGP.

Lesson 4.2: IP Version 4 Addressing

IPv4 Address Format

- IPv4 addresses are 32-bit numbers, usually written in dotted decimal format (e.g., 192.168.1.1).
- Addresses consist of network and host portions.

Network Masks

- Used to identify the network portion of an IP address.
- Valid masks have contiguous bits set to 1, followed by bits set to 0.
- Prefix notation (slash notation) indicates how many bits are in the network mask (e.g., /24).

Subnet Masks

- Help divide a network into smaller subnetworks.
- Common subnet mask bits and their decimal equivalents:
 - 1 bit: 128
 - 2 bits: 192
 - 3 bits: 224
 - 4 bits: 240
 - 5 bits: 248
 - 6 bits: 252
 - 7 bits: 254
 - 8 bits: 255

Host Address Ranges

- Range of valid IP addresses for devices within a subnet.

Default Gateway

- The device that routes traffic from a local network to other networks.

Broadcast Addresses

- Used to send packets to all hosts on a subnet.

IP Interface Configuration

- Windows: Static vs dynamic (DHCP), adapter naming, configuration methods.
- Linux: Interface identification, running vs persistent configuration, configuration methods.

Lesson 4.3: IP Version 4 Subnetting

Classful Addressing

- Divides IPv4 into classes A, B, C, D, and E based on address ranges.

Public vs Private Addressing

- **Public IPs:** Routable on the internet, managed by IANA.
- **Private IPs:** Reserved for internal use, non-routable on the internet.
 - Class A: 10.0.0.0 to 10.255.255.255
 - Class B: 172.16.0.0 to 172.31.255.255
 - Class C: 192.168.0.0 to 192.168.255.255

Other Reserved Address Ranges

- Class D: 224.0.0.0 to 239.255.255.255 (Multicast)
- Class E: 240.0.0.0 to 255.255.255.255 (Experimental)
- Loopback: 127.0.0.0 to 127.255.255.255

IPv4 Address Scheme Design

- Planning considerations for subnetting.
- Rules for subnet and host calculations.
- Designing subnets to optimize network efficiency.

Classless Inter-Domain Routing (CIDR)

- Allows more flexible allocation of IP addresses than classful addressing.

Variable Length Subnet Masks (VLSM)

- Enables subnetting a network into different sized subnets for efficient IP use.

Lesson 4.4: IP Troubleshooting Tools

Common Tools

- **ipconfig**: Windows tool for viewing and managing IP configuration.
- **ifconfig and ip**: Linux tools for network configuration.
- **arp**: Displays and modifies IP to MAC address mappings.
- **ping**: Tests connectivity using ICMP echo requests.

ipconfig Commands

- `ipconfig`
- (basic)
- `ipconfig /all`
- (detailed info)
- `ipconfig /renew`
- ,
- `/release`
- (DHCP control)
- `ipconfig /flushdns`
- (clear DNS cache)
- `ipconfig /registerdns`
- (register DNS)

ifconfig and ip Commands

- `ifconfig`
- (deprecated but still used)
- `ip addr`
- ,
- `ip link`
- ,
- `ip -s link`
- (modern commands for IP and interface info)

arp Commands

- `arp -a`
- : Show ARP table.
- `arp -s`
- : Add static ARP entry.
- `arp -d *`
- : Delete all ARP entries.
- `ip neigh`
- : Show neighbor table.

ping Usage

- Basic ping to test connectivity.
- Common errors: Destination host unreachable, Request timed out.
- Ping switches for advanced use.

Lesson 4.5: IP Version 6

IPv4 vs IPv6

- IPv6 uses 128-bit addresses vs 32-bit in IPv4.
- Designed to mitigate address exhaustion.
- Different notation and header structure.

IPv6 Address Format

- Hexadecimal colon-separated notation.
- Compression rules: canonical notation, double colon (::) usage.

IPv6 Network Prefixes

- Specify network portion in IPv6 addresses.

IPv6 Unicast Addressing

- Global unicast and Interface ID (EUI-64).

IPv6 Link Local Addressing

- Automatically configured addresses for local network communication.

IPv6 Multicast and Anycast Addressing

- Multicast uses protocols like MLD.
- No broadcast in IPv6.
- Neighbor Discovery (ND) protocol replaces ARP.
- Anycast used for nearest-node communication.

IPv4 and IPv6 Transition Mechanisms

- Dual stack, tunneling, and NAT64 to support coexistence.

Common IPv6 Address Prefixes

- Global unicast: 2000::/3
- Link local: fe80::/10
- Multicast: ff00::/8
- Loopback: ::1/128
- Documentation examples: 2001:db8::/32

Lesson 4.6: IP Troubleshooting

Common Issues

- IP configuration errors: wrong IP address or subnet mask.
- Duplicate IP and MAC addresses:
 - Can cause conflicts and connectivity loss.
 - Diagnosed with arp, ipconfig, ip neigh.
- IP forwarding issues:
 - Incorrect routing can prevent packet delivery.

Module Summary

- IPv4 addressing uses 32-bit addresses to uniquely identify devices.
- IPv4 forwarding routes packets based on destination IP and routing tables.
- Configuring IP networks involves assigning IPs, subnet masks, gateways, and subnetting for efficiency.
- Tools like ping, ipconfig/ifconfig, and tracert help test and troubleshoot IP configurations.
- IPv6 uses 128-bit addressing to support vast addressing needs with features like hierarchical design.
- Troubleshooting involves verifying configurations, diagnosing conflicts, and using network tools.

Vocabulary

- **Address Resolution Protocol (ARP):** A protocol used to map an IP address to a physical machine (MAC) address on a local network.
- **Anycast Addressing:** A network addressing method where data is routed to the nearest or best destination out of a group of potential receivers.
- **Broadcast Addressing:** Sending data to all devices on a local network segment.
- **Classful Addressing:** The original system dividing IP addresses into fixed classes (A, B, C, D, E) based on address ranges.
- **Classless Inter-Domain Routing (CIDR):** A method for allocating IP addresses and routing that replaces the classful system with flexible prefix lengths.
- **Default Gateway:** A network device that routes traffic from a local network to other networks or the internet.
- **DHCP (Dynamic Host Configuration Protocol):** A protocol that automatically assigns IP addresses and other network configuration to devices.
- **Duplicate IP Address:** When two devices on the same network have the same IP address, leading to conflicts.
- **EUI-64:** A method for automatically generating an IPv6 Interface ID using a device's MAC address.
- **Flow Label:** A field in the IPv6 header used to identify packets requiring special handling.
- **Host Address Range:** The set of valid IP addresses that can be assigned to devices on a subnet.
- **ICMP (Internet Control Message Protocol):** Used by network devices to send error messages and operational information.
- **Interface ID:** The portion of an IPv6 address that identifies a specific interface on a subnet.
- **IPv4 (Internet Protocol version 4):** The fourth version of the Internet Protocol, using 32-bit addresses.
- **IPv6 (Internet Protocol version 6):** The newest Internet Protocol version, using 128-bit addresses.
- **Link Local Addressing:** IPv6 addresses automatically assigned for communication within the local network segment.
- **Loopback Address:** An IP address (127.0.0.1 for IPv4, ::1 for IPv6) used by a host to send packets to itself.
- **MAC Address (Media Access Control Address):** A hardware address that uniquely identifies each device on a local network.
- **Multicast Addressing:** Sending data to a group of interested receivers.
- **Network Mask / Subnet Mask:** A binary mask used to identify the network portion of an IP address.

- **Neighbor Discovery (ND):** Protocol in IPv6 that replaces ARP to find other devices on the local network.
- **Prefix (Slash) Notation:** A shorthand for writing IP network masks indicating the number of bits used for the network.
- **Private IP Address:** IP addresses reserved for use within private networks, not routable on the internet.
- **Public IP Address:** IP addresses that are routable on the internet.
- **Routing Table:** A data table stored in a router or host that lists routes to particular network destinations.
- **Subnet:** A logically visible subdivision of an IP network.
- **Subnetting:** The process of dividing a network into smaller subnetworks.
- **Tunneling:** A method used to send IPv6 packets over an IPv4 network.
- **Variable Length Subnet Mask (VLSM):** Allows subnets to have different sizes by using different subnet masks.
- **Unicast Addressing:** One-to-one communication between a single sender and a single receiver.

Acronyms

Acronym	Meaning
ARP	Address Resolution Protocol
BGP	Border Gateway Protocol
CIDR	Classless Inter-Domain Routing
DHCP	Dynamic Host Configuration Protocol
EUI-64	Extended Unique Identifier (64-bit)
ICMP	Internet Control Message Protocol
IGMP	Internet Group Management Protocol
IP	Internet Protocol
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6

MAC	Media Access Control
MLD	Multicast Listener Discovery
ND	Neighbor Discovery
NIC	Network Interface Card
VLSM	Variable Length Subnet Mask

